

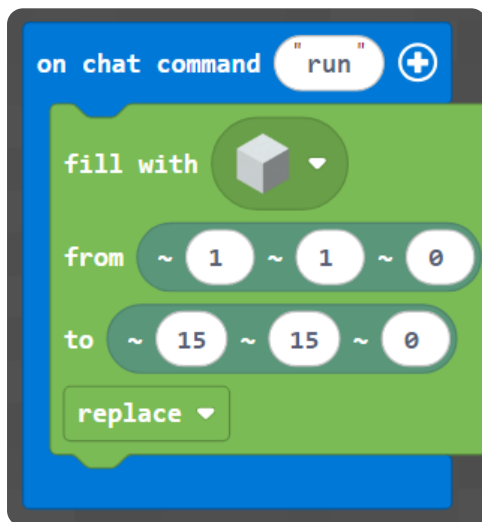
M003 Week 3 — English Worksheet (Advanced)

Topic: Pixel Art on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 45 minutes

This week you copy a picture made of blocks onto your wall. Every block sits at a spot named by **two numbers** — (x, y). x is how far **across**, y is how far **up**. You read the picture, work out each block's (x, y), and place it. The harder part is reading several blocks at once and seeing the pattern they make together.

First, build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot** — not from the corner of the world. Stand on the red block *every* time you run, or the whole picture shifts. Move your feet, move your picture.

1 · Predict 🧠

Read each set of steps. Do not run them. For each one, write **what you think you will see** and **why** you think so.

```
place red block at (5, 1)
place red block at (6, 1)
place red block at (7, 1)
```

Across or up? Which number stays the same, and why does that decide the direction?


```
place green block at (8, 13)
place green block at (8, 14)
place green block at (8, 15)
```

Near the top or the bottom of the wall? Across or up? Explain how you know.


```
place yellow block at (6, 1)
place yellow block at (7, 2)
place yellow block at (8, 3)
```

Both numbers grow by 1 each time. Row, column, or a stair going up? Why?


```
place yellow block at (5, 1)
place yellow block at (6, 1)
place yellow block at (5, 2)
place yellow block at (6, 2)
```

Four blocks together. What shape do they make, and where does it sit? Explain your reason.

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what it is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — These should make the **bottom row of the pineapple**: five yellow blocks from (6, 1) to (10, 1). Right now they all land in the **same spot**.

```
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The apple stem should go **up** at $x = 8$, at (8, 13), (8, 14), (8, 15). Right now it grows **across** instead.

```
place brown block at (13, 8)
place brown block at (14, 8)
place brown block at (15, 8)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — A friend stood on the **wrong block** and ran code that matched the grid exactly. The picture came out, but every block landed at an x **one bigger** than the grid says — the whole picture shifted across. The code itself is fine.

```
(the code matched the grid exactly)
```

What did your friend forget to do before pressing run? Why does the home spot change where (x, y) lands, even when the code is correct?

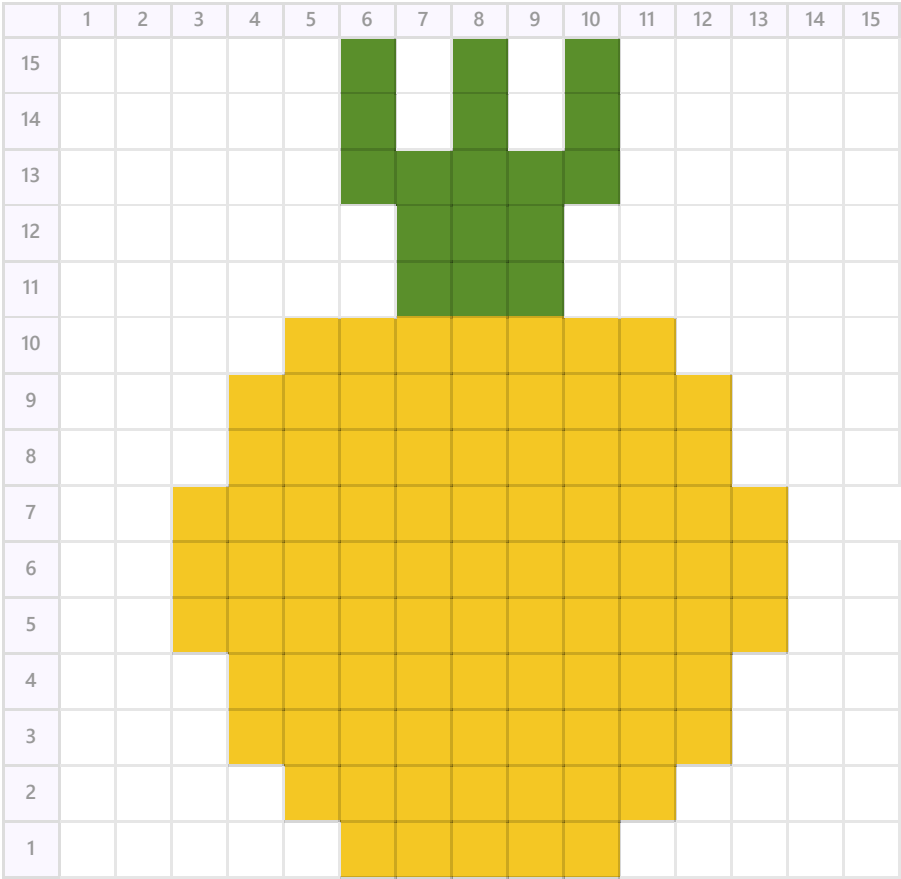
3 · Build It 📷

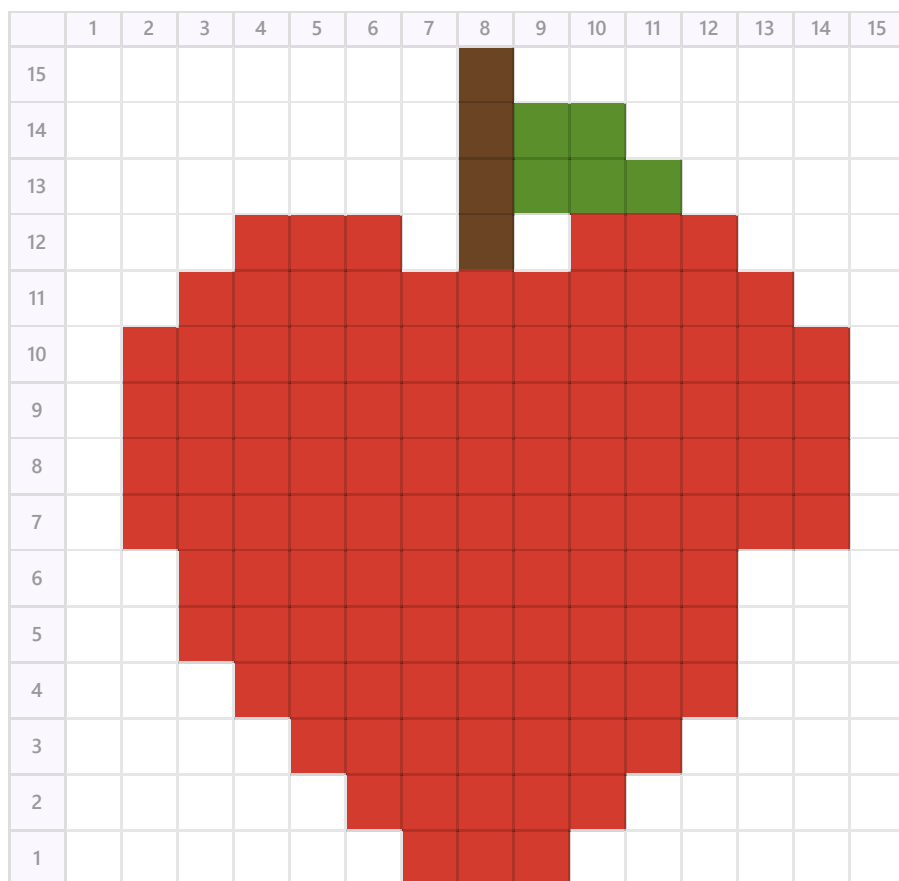
Now switch to your homework world and stand on your **home spot** (the red block). The numbers across the **top** are **x**, the numbers down the **side** are **y**.

Plan first. Before you place anything, look at the 🍍 pineapple and write the (x, y) of its **bottom-left** block and its **top** block. This tells you how wide and how tall it is.

Now copy a picture below onto your wall, block by block, by reading the (x, y) of every colored square.

🍍 Pineapple





Modify challenge. After you finish one fruit, change something of your own — swap a colour, add a leaf, or make the picture one block bigger. Write the (x, y) of the blocks you changed or added.

When your picture is finished, send a photo, then explain what you did. Use these sentence starters — write 5 to 6 sentences total.

First, I chose to build the ...

The lowest block in my picture is at $(x = \dots, y = \dots)$.

To go across a row I changed the ... number; to go up I changed the ... number.

The trickiest part to read off the grid was ...

For my modify challenge, I changed ... at (x = ..., y = ...).

If I had more time, I would ...

4 · Explain It

Now take a video on your phone while you show your pixel art on the wall. Talk through it like you are teaching someone who has never seen it. Try to use these words: **x, y, coordinate, across, up, home spot**.

1. Show your home spot (the red block) and say why you stand on it every time.
2. Show your finished picture from the front.
3. Point at one block and read its coordinate out loud: "this one is at x ..., y ...".
4. Say in your own words how (x, y) decides exactly where a block lands.

Write a short paragraph (5 to 6 sentences) about what you built and what you learned. Then answer: if you were teaching a friend who had never done this, what would you show them first, and why?

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.